

Virtual Memory, Paging Architecture & Page Replacement

Subject: CS401 | Unit 3: Memory Management

1. Executive Overview & Syllabus Scope

Covers Address Translation, TLB hits/misses, Multi-level Paging, Inverted Page Tables, FIFO, LRU, Optimal Page Replacement

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Memory Management & Virtual Memory Architecture
- 1. Logical vs Physical Address Translation: Page Number p and Page Offset d
- 2. Translation Lookaside Buffer TLB: Associative lookup and Effective Access Time EAT
- 3. Multi-Level & Inverted Page Tables: Structure, scalability, and memory overhead tradeoffs
- 4. Page Replacement Algorithms: FIFO, LRU Least Recently Used, Optimal Belady's Min
- 5. Belady's Anomaly: Why increasing frames can increase page faults in FIFO
- 6. Thrashing: Degree of multiprogramming collapse and Working-Set Model prevention

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.